DATA VISUALIZATION · NYU

Equity-Safety Gap Analyzer

Visualizing where Vision Zero investments miss the neighborhoods that need them most

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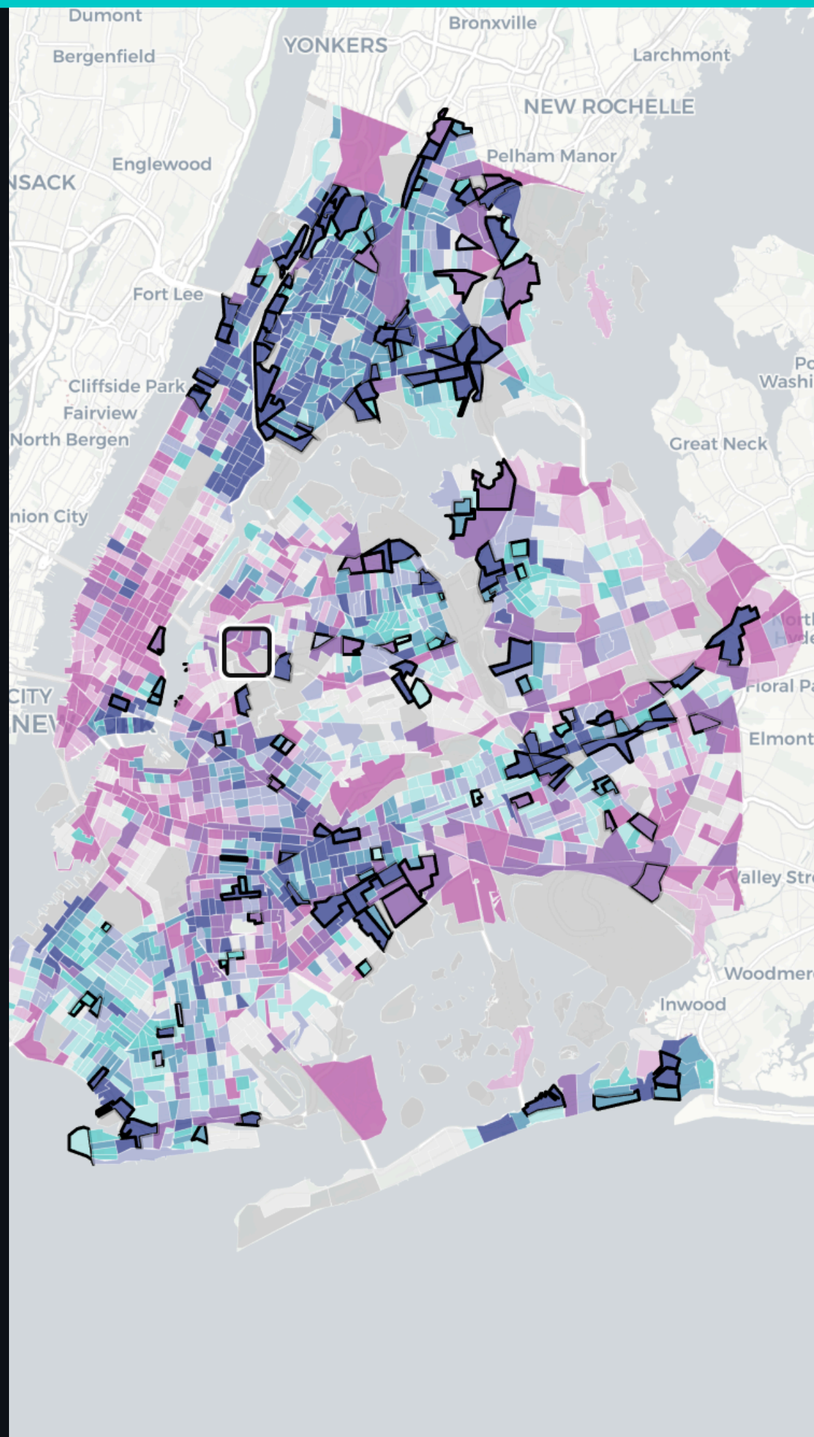
2,325 Census Tracts

16 SVI Variables

1,200+ VZ Projects



The Problem

Vision Zero came out in 2014

NYC promised zero traffic deaths.
1,200+ street improvements.
Speed limit cut to 25 mph.
10 years of sustained investment.

Aggregate Numbers Look Good

Overall traffic deaths declined.
DOT 2024 report: streets are
"safer and more equitable."
Headlines are positive.

But Equity Tells Another Story

Transportation Alternatives (2024):
fatalities INCREASED in majority
Black & Brown neighborhoods
over the same decade.

Literature: Abebe et al. (2024) Transport Policy 149 · Dragan & Glied (2024) AJPH 114(6) · Cui et al. (2025) AAP · Dumbaugh & Stiles (2025) JTLU

Research Framework

Where in NYC do high crash risk + high social vulnerability co-occur, yet receive little Vision Zero investment?

RISK LAYER

NYPD Crashes 2014-2024

KSI = Killed ×3 + Injured ×1
Min-max normalised per tract
→ $\hat{R}_i$

INVESTMENT LAYER

NYC DOT Vision Zero View

LPIs · Speed Humps · SIPs
Bike lane miles · Weighted
→ $\hat{I}_i$

EQUITY LAYER

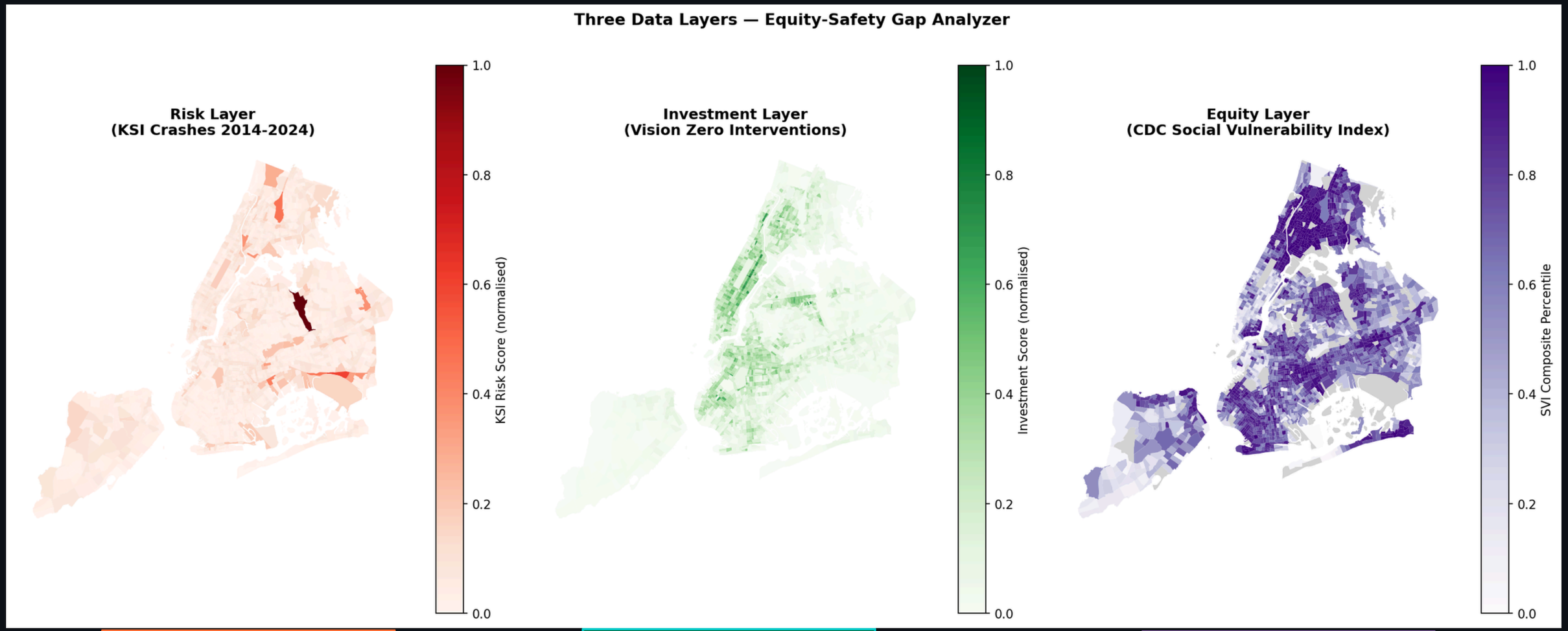
CDC/ATSDR SVI 2022

16 census variables, 4 themes
Composite percentile 0 → 1
→ SVI_i

Gap Score = $\hat{R}_i - \hat{I}_i$ **Safety Desert: Gap > 0 AND SVI ≥ 0.75** (Li et al. 2022; Kasha et al. 2025; Strode et al. 2020)

Three Data Layers

Each layer independently tells part of the story. The project intersects all three at the census-tract level.

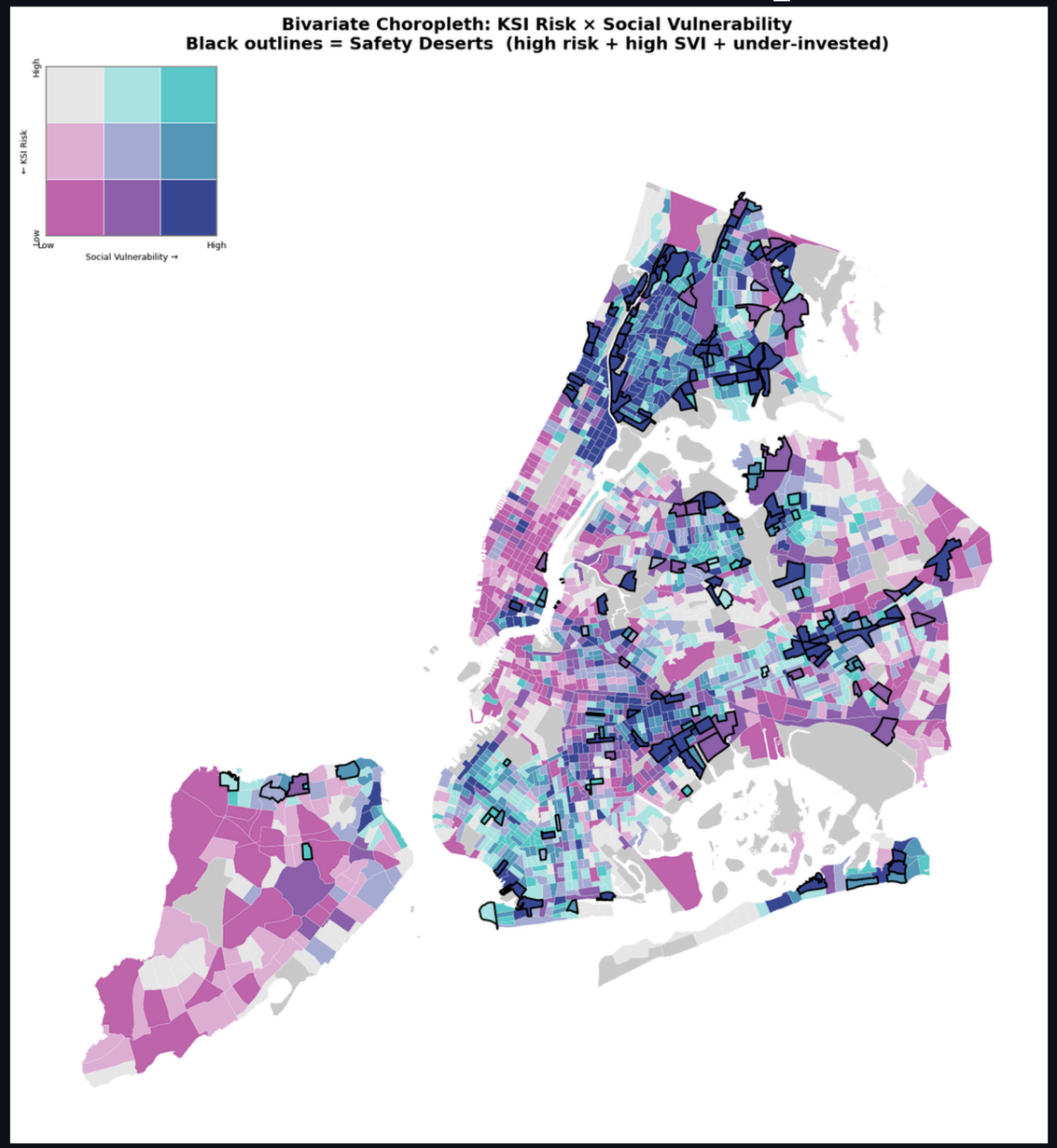


RISK (KSI Crashes)

INVESTMENT (VZ Interventions)

EQUITY (SVI)

Bivariate Choropleth Map



Crisis Zones

Deep purple = high KSI AND high SVI. Concentrated in South Bronx, East New York, SE Queens.

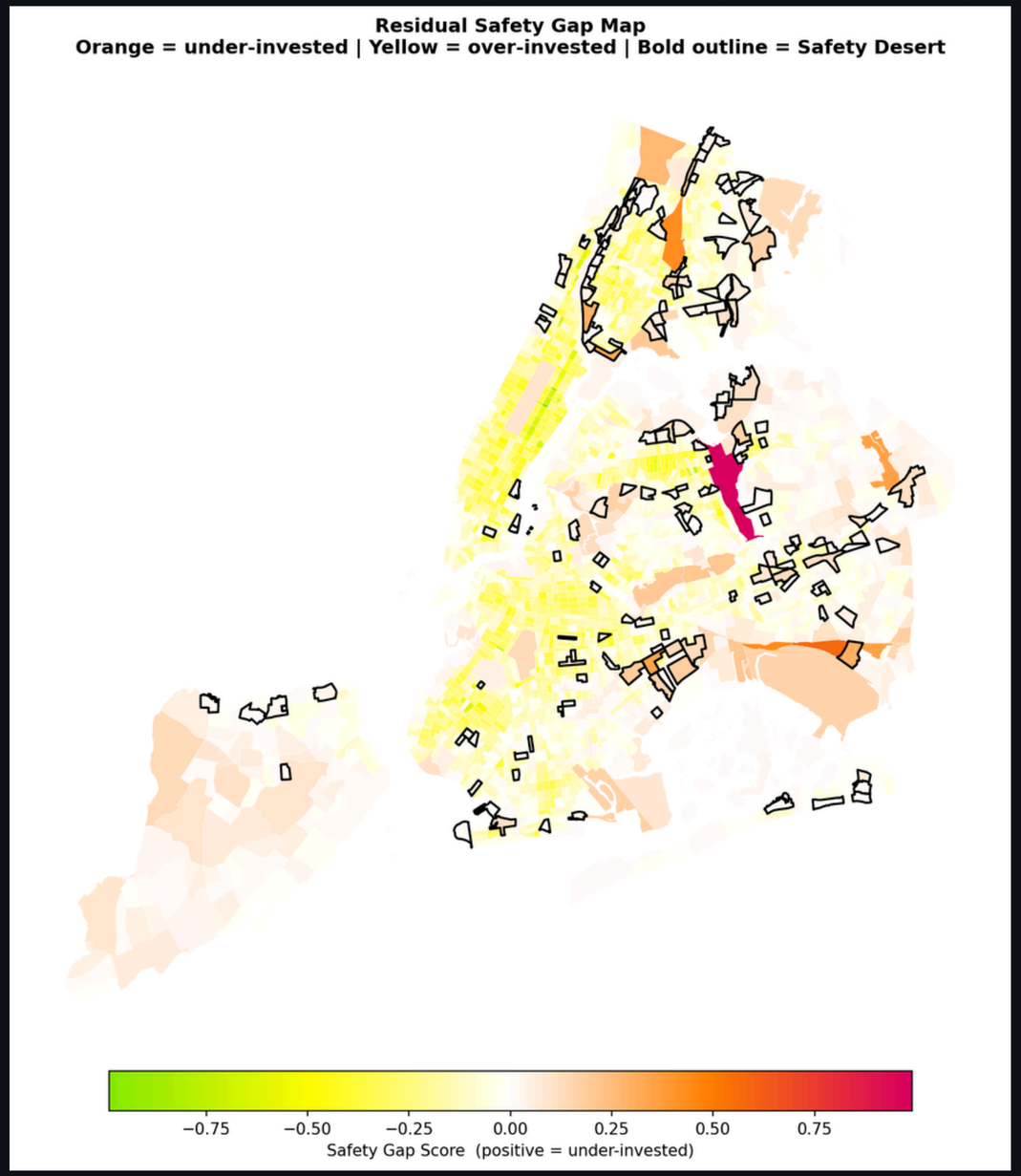
Colour Grid Logic

X-axis encodes SVI (yellow→blue). Y-axis encodes KSI risk (yellow→red). High-high = deep purple.

Black Outlines

Safety Deserts: high risk + high SVI + under-invested vs. investment score. Key policy targets.

The Safety Gap Map



How to Read This Map

Gap = $\hat{R}_i - \hat{I}_i$

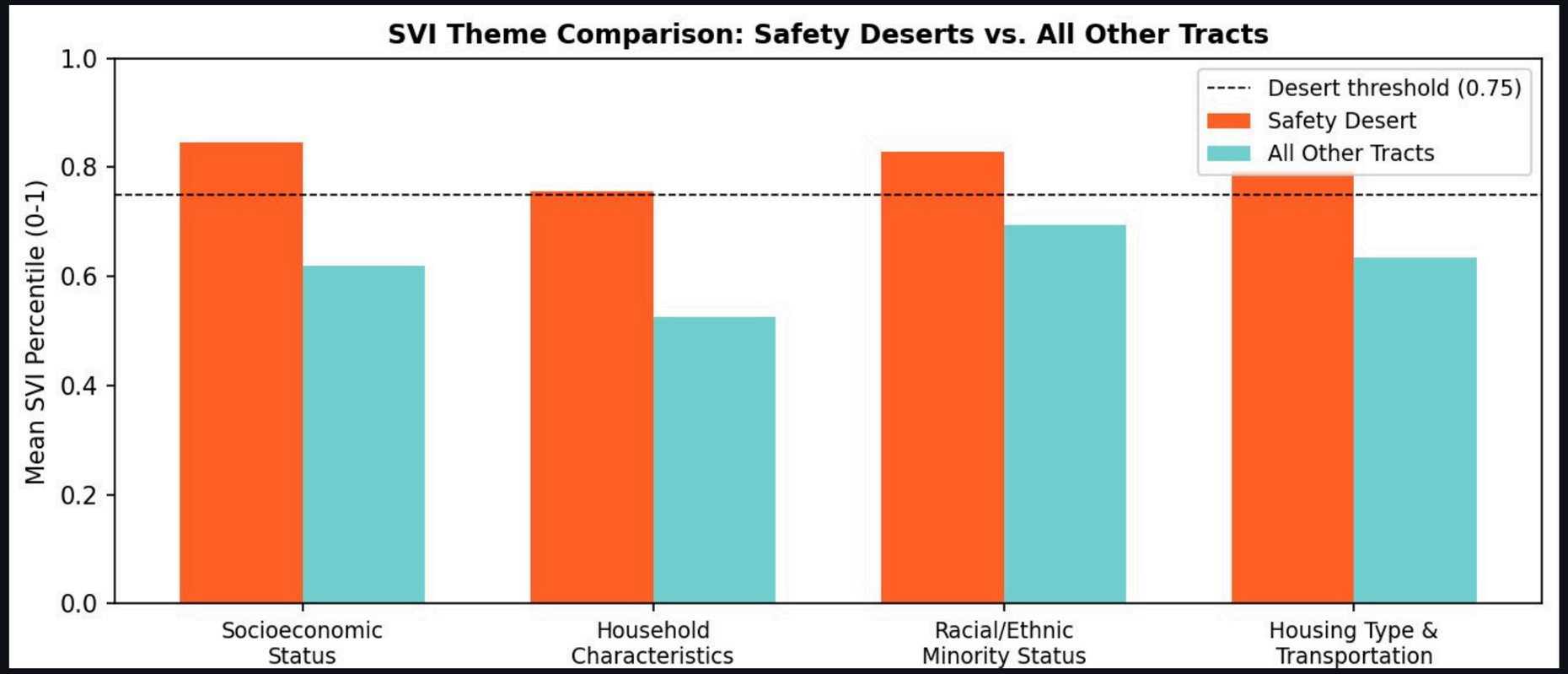
Pink→ risk outpaces investment
Yellow→ investment exceeds risk
Bold outlines → Safety Desert

Key Findings

The Bronx shows the highest density of Safety Desert outlines. Manhattan's bike-lane corridor is strongly teal -- over-invested relative to crash severity. Staten Island and outer Queens are broadly orange. This pattern directly contradicts the framing that Vision Zero is equitably distributed.

Equity Analysis

SVI Theme Comparison

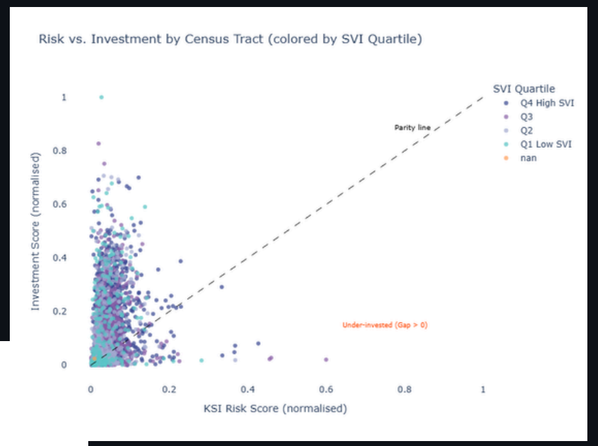


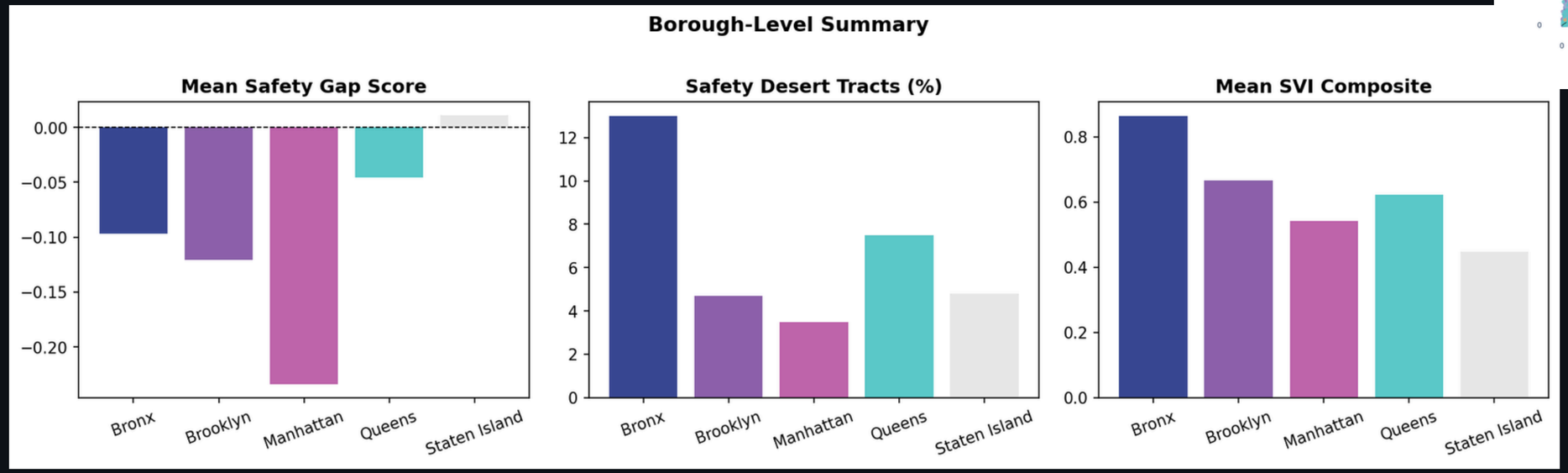
Socioeconomic gap is biggest

Safety Deserts score 0.85 vs 0.62 on socioeconomic SVI -- a 37% gap. This is the most discriminating dimension.

Equity Analysis

Borough Summary





Bronx leads Safety Deserts

~13% of Bronx tracts are Safety Deserts -- highest of any borough, consistent with Abebe et al. 2024.

Manhattan is over-invested

Negative mean gap score -- bike-lane density far exceeds crash severity relative to the Bronx or Queens.

Dashboard

Gap Score - Demo!!

So What?

Vision Zero succeeds on average -- but averages hide who gets left behind.

Bivariate Maps Work

Encoding two risk dimensions simultaneously lets viewers spot Crisis Zones instantly -- no tables or regression output required.

Safety Deserts Are Real

Bronx tracts with high SVI and high crash rates are systematically under-served. This is now visible and quantifiable.

Investment Bias

Manhattan's bike-lane corridor shows the highest investment density despite lower relative crash severity than the Bronx.

Actionable Output

GeoJSON + Folium dashboard give planners a tract-level tool to prioritise the next round of Vision Zero interventions.